MULTI-PURPOSE AGRICULTURE DRONE

PROJECT NAME: Multi-Purpose Agriculture Drone

PROJECT GROUP: CLASS OF 2024

WHY THIS PROJECT ?

- Manual spraying is time-consuming, inefficient, and labor-intensive.
- Uneven distribution leads to wastage of water, seeds, or fire retardants.
- Hard-to-reach or hazardous areas are difficult to cover safely.
- Drones provide precise, fast, and safe spraying for agriculture, reforestation, and firefighting.

INTRODUCTION

- This project focuses on using drones for spraying water, seeds, or fire retardants efficiently over large or hard-to-reach areas.
- It reduces manual labor, improves precision, and ensures safer and faster spraying operations.

KEY FEATURES

- Efficient spraying of water, seeds, fertilizer, or flower distribution
- Covers plants, flowers, large fields, and hard-to-reach areas
- Reduces manual labor and effort
- Ensures precise and uniform distribution
- Supports safe and faster operations

ARCHITECTURE

- Carbon fiber Hexacopter frame design for stability
- BLDC motors : HOBBYWING X8 Motor
- Flight Controller : Jiyi K++ V2
- Li-Po Battery : Pair of Tattu 22000mAh

RESULTS

- Drone successfully performed spraying operations for water, seeds, fire retardants, and flowers.
- Materials were distributed evenly and precisely over the target area.
- Reduced manual labor and increased efficiency in spraying tasks

FUTURE SCOPE

1. **Integration of AI-based crop analysis**
2. **Autonomous navigation using GPS**
3. **Multi-payload system for spraying and monitoring**